

FOND DU LAC, WI, USA

WMO: 726506

Lat: 43.769N

Lon: 88.491W

Elev: 807

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-1.7	-15.5	2.4	-5.4	-10.9	3.1	-1.3	24.8	18.4	22.8	20.1	7.2	290	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	88.5	73.4	85.7	71.6	83.0	70.2	76.2	85.1	74.3	82.6	72.5	80.4	11.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	126.3	82.1	71.7	120.9	80.3	69.9	113.4	78.0	40.1	84.4	38.3	82.7	36.6	80.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.2	18.3	-13.2	92.7	5.6	3.3	-17.3	95.0	-20.6	96.9	-23.7	98.7	-27.8	101.1
			-13.8	79.2	5.5	2.4	-17.7	80.9	-21.0	82.2	-24.0	83.6	-28.0	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	19.5	22.6	33.5	46.0	57.7	67.4	71.8	69.4	62.6	50.2	37.6	25.4	
	DBStd	20.44	12.06	11.51	11.57	9.13	8.61	6.71	5.70	5.28	7.93	9.09	9.61	11.24	
	HDD50	3714	946	768	529	181	24	0	0	0	5	112	386	763	
	HDD65	7144	1411	1187	977	573	260	51	10	19	138	469	823	1226	
	CDD50	2651	0	1	18	60	262	521	675	602	382	117	12	1	
	CDD65	607	0	0	1	2	33	122	219	156	65	9	0	0	
	CDH74	5115	0	0	8	40	334	1051	1949	1142	518	73	0	0	
	CDH80	1411	0	0	0	5	90	293	618	273	121	11	0	0	
Wind	WSAvg	8.6	9.3	9.3	9.4	10.1	9.1	7.9	7.1	6.5	7.5	8.7	9.4	9.1	
Precipitation	PrecAvg	29.1	1.1	0.9	1.7	2.6	3.2	3.6	3.5	3.6	3.6	2.3	1.9	1.3	
	PrecMax	39.1	3.8	2.2	5.0	5.9	13.5	7.2	8.1	8.1	12.7	7.0	5.7	4.1	
	PrecMin	22.6	0.0	0.1	0.1	0.9	0.5	0.3	0.9	0.4	0.2	0.2	0.0	0.0	
	PrecStd	4.6	0.7	0.6	1.1	1.1	2.2	1.7	1.6	1.9	2.6	1.5	1.3	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	55.1	72.8	79.5	88.0	91.0	93.7	89.8	88.1	81.4	67.8	57.1	
		MCWB	43.4	49.0	60.9	61.5	70.6	72.9	76.6	75.2	71.7	68.0	56.5	53.8	
	2%	DB	41.7	46.4	63.3	72.4	82.0	86.5	89.6	85.9	83.5	74.9	61.2	48.6	
		MCWB	37.5	40.9	54.9	57.5	66.6	71.7	74.7	72.9	70.2	63.8	53.3	45.1	
	5%	DB	37.8	41.4	55.9	66.1	77.4	83.4	86.0	82.7	80.3	70.3	56.6	43.4	
		MCWB	34.9	37.3	48.4	54.5	63.5	70.0	72.2	70.6	67.6	61.1	50.1	39.6	
	10%	DB	35.9	37.1	49.8	61.8	72.8	80.6	83.0	80.7	76.5	64.5	52.2	39.3	
		MCWB	33.7	33.9	44.1	50.8	61.6	68.1	71.0	69.2	65.8	57.2	46.8	36.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	49.1	62.3	64.2	72.3	77.0	79.7	77.6	74.8	70.3	58.7	54.1
			MCDB	47.0	54.5	72.4	74.2	84.5	87.7	90.0	86.3	83.6	78.6	65.2	56.9
2%		WB	38.1	41.5	56.4	59.8	68.5	74.4	76.9	75.0	72.0	66.4	54.6	45.9	
		MCDB	40.9	45.9	62.0	69.8	78.1	83.7	86.1	83.0	80.0	72.5	59.8	47.8	
5%		WB	35.4	37.7	49.7	56.0	66.0	72.1	74.9	72.8	70.0	62.1	50.9	40.6	
		MCDB	37.7	41.4	55.3	65.2	75.0	80.6	83.0	80.4	77.2	68.4	55.9	43.0	
10%		WB	33.3	34.4	43.9	51.8	63.2	69.5	73.0	70.8	67.8	58.2	46.5	36.6	
		MCDB	35.5	37.1	49.7	60.6	70.8	77.4	80.7	77.8	74.6	64.6	51.2	39.4	
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.0	16.3	18.7	19.6	19.0	19.5	18.5	19.9	17.4	14.8	13.1	
		MCDBR	14.0	16.3	22.5	26.7	24.8	22.2	21.7	20.8	22.6	23.4	21.2	15.5	
		MCWBR	12.0	12.5	15.3	16.0	13.5	11.8	10.6	10.6	11.7	14.1	15.1	12.9	
	5% WB	MCDBR	13.6	15.4	21.1	24.5	21.8	19.7	18.7	17.9	19.2	20.2	19.6	14.4	
		MCWBR	12.0	12.2	15.5	16.2	13.1	12.1	10.8	10.4	11.3	14.3	15.6	13.1	
Clear-Sky Solar Irradiance	taub		0.275	0.320	0.338	0.380	0.404	0.417	0.408	0.402	0.372	0.334	0.298	0.276	
	taud		2.388	2.240	2.348	2.279	2.263	2.265	2.294	2.301	2.378	2.477	2.507	2.436	
	Ebn at Noon		273	273	285	280	276	271	272	270	270	268	263	262	
	Edh at Noon		24	34	35	40	42	42	41	39	33	26	22	21	
All-Sky Solar Radiation	RadAvg		511	849	1205	1444	1718	1903	1942	1662	1341	852	547	392	
	RadStd		68	87	135	136	137	137	89	106	109	93	46	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page